

Problem Solving and Testing Using Java :

Session 1:

A-JAVA COMPARATOR :

```
import java.util.*;

class Checker implements Comparator<Player> {

    public int compare(Player a, Player b) {

        if (a.score != b.score) {

            return b.score - a.score;

        }

        return a.name.compareTo(b.name);

    }

}

class Player{

    String name;

    int score;

    Player(String name, int score){

        this.name = name;

        this.score = score;

    }

}

class Solution {

    public static void main(String[] args) {

        Scanner scan = new Scanner(System.in);

        int n = scan.nextInt();

        Player[] player = new Player[n];

        Checker checker = new Checker();

        for(int i = 0; i < n; i++){

            player[i] = new Player(scan.next(), scan.nextInt());

        }

        scan.close();

        Arrays.sort(player, checker);
```

```
for(int i = 0; i < player.length; i++){  
    System.out.printf("%s %s\n", player[i].name, player[i].score);  
}  
}  
}
```

OUTPUT:

✓ Test case 0	1 5 2 amy 100 3 david 100 4 heraldo 50 5 aakansha 75 6 aleksa 150
✓ Test case 1	
✓ Test case 2	
✓ Test case 3	
✓ Test case 4	
✓ Test case 5	

Expected Output

1 aleksa 150
2 amy 100
3 david 100
4 aakansha 75
5 heraldo 50

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B - Java Lambda Expressions :

```
import java.io.*;

import java.util.*;

interface PerformOperation {

    boolean check(int a); }

class MyMath {

    public static boolean checker(PerformOperation p, int num) {

        return p.check(num);

    }

    PerformOperation isOdd() {

        return n -> n % 2 != 0;

    }

    PerformOperation isPrime() {

        return n -> {

            if (n < 2) return false;

            for (int i = 2; i * i <= n; i++) {

                if (n % i == 0)

                    return false;

            }

            return true;

        };

    }

    PerformOperation isPalindrome() {

        return n -> {

            int temp = n, rev = 0;

            while (temp > 0) {

                rev = rev * 10 + temp % 10;

                temp /= 10;

            }

            return rev == n;

        };

    }

}
```

```

    }
}

public class Solution {

    public static void main(String[] args) throws IOException {

        MyMath ob = new MyMath();

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

        int T = Integer.parseInt(br.readLine());

        PerformOperation op;

        boolean ret = false;

        String ans = null;

        while (T--> 0) {

            String s = br.readLine().trim();

            StringTokenizer st = new StringTokenizer(s);

            int ch = Integer.parseInt(st.nextToken());

            int num = Integer.parseInt(st.nextToken());

            if (ch == 1) {

                op = ob.isOdd();

                ret = ob.checker(op, num);

                ans = (ret) ? "ODD" : "EVEN";

            } else if (ch == 2) {

                op = ob.isPrime();

                ret = ob.checker(op, num);

                ans = (ret) ? "PRIME" : "COMPOSITE";

            } else if (ch == 3) {

                op = ob.isPalindrome();

                ret = ob.checker(op, num);

                ans = (ret) ? "PALINDROME" : "NOT PALINDROME"; }

            System.out.println(ans);

        }

    }

}

```

OUTPUT :

✓ Test case 0

✓ Test case 1 

✓ Test case 2 

1	5
2	1 4
3	2 5
4	3 898
5	1 3
6	2 12

Expected Output

1	EVEN
2	PRIME
3	PALINDROME
4	ODD
5	COMPOSITE

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